LEARNMATE AI — LEAN CANVAS (SOCIAL IMPACT & PRODUCT ARCHITECTURE)

BharatCares Machine Learning & Applied AI Internship Project | UN SDG 4: Quality Education | Candidate: Joel Francis (B.Tech CSE)

1. Problem
• Students struggle to identify precise skill gaps across CS & AI topics.
• Lack of continuous study discipline and high dropout rates in self-paced learning.
• One-on-one personal tutoring is prohibitively expensive for underserved students.
Existing Alternatives: Static YouTube tutorials, generic LeetCode drills, paid LLM tools with hallucination risks.

4. Solution
• **Diagnostic Assessments:** Interactive timed quizzes with instantaneous explanations.
• **ML Support Engine:** Random Forest classifying students into 3 actionable support tiers.
• **Adaptive Study Planner:** Milestone schedules prioritizing diagnosed weak areas.
• **AI Academic Tutor:** Local TF-IDF semantic retrieval over verified knowledge base.

8. Key Metrics
• Diagnostic quiz completion rate (>75%).
• Weekly learning streak retention (>7 days).
• Pre-to-post test score improvement (+15 to 20%).
• Task checklist completion rate in adaptive planner.

3. Unique Value Proposition
• **Your Personal AI Learning and Student Success Companion.**
• Transparent Scikit-Learn ML classification (88.3% accuracy) diagnosing student support needs.
• Orchestrated 5-agent local AI collective (Tutor, Quiz, Analyst, Planner, Career) running offline without paid API costs.

9. Unfair Advantage
• Fully functional, verifiable local multi-agent orchestration architecture.
• Complete model transparency (no black-box or hallucinated credentials).
• Purpose-built for UN SDG 4 social impact with zero paid software dependencies.

5. Channels
• Direct academic university partnerships & college training cells.
• BharatCares community learning centers and tech NGOs.
• Open-source GitHub repository and lightweight offline-first web platform.

2. Customer Segments
• **Primary:** Undergraduate Engineering & Computer Science students (Tier-2/3 institutions).
• **Secondary:** Self-taught coders, AI bootcamp participants, and vocational learners.
• **Early Adopters:** 3rd & 4th-year students preparing for technical placements & exams.

7. Cost Structure
• Zero ongoing LLM API expenses (local deterministic & TF-IDF models).
• Low-cost lightweight cloud hosting (AWS EC2 / DigitalOcean ~$15/month).
• Curriculum verification, software maintenance, and domain expert review.

6. Revenue & Sustainability Model
• **Current:** 100% Free & Open-Source academic prototype for social good (SDG 4).
• **Future B2B:** Institutional analytics licensing for colleges to monitor batch performance.
• **Future B2C:** Freemium tier for personalized mock technical interview simulations.